

automated fare collection system in India. Currently in the third phase of the Metro expansion project, it is working with Thales Group for deploying more AFC systems in the new stations. A total of 895 entry and exit terminals, 485 ticket vending machines and 126 recharge terminals are in the works for installation.

However, AFC applications are not limited to transportation alone. According to a report from AFC Systems, a company manufacturing automated fare collection solutions, the AFC market will grow to about \$12 billion by 2021 with applications in museums, theme parks and retail witnessing exponential rise.

The investment

The cost of an automated fare system depends on the scale of usage. For example, fare collection system for smaller businesses like individual theatres, food joints and private libraries require a smaller setup and investment. Bus and taxi transportation service providers require handheld collection devices, while subways and train facilities require much bigger infrastructure.

According to AFC Systems, "The price is based on multiple factors and related to the scale of integration. For example, if you want to implement AFC solutions in 25 buses, the cost will be ₹ 1.5 million to ₹ 2 million." On the other hand, subway projects are much costlier and mainly work on tendered quotations. For example, the installation cost of AFC setup at the Gurgaon metro rail project was ₹ 200 million, while the Pune Metro AFC project cost around ₹ 520 million without taxes.

As regards monetary benefits, AFC Systems shared, "Business can recover ROI based on the investment involved and transportation area. The ROI can be achieved within five to seven years." **EFY**

—Paromik Chakraborty, technical journalist, EFY

Solar ROI for IIT Roorkee And Delhi International Airport

► While the IIT Roorkee's solar setup cuts its electricity bill by over 15 per cent, DIAL is saving up to 20 per cent

Electricity costs constitute a substantial chunk of the complete operational expenditure of organisations. Solar system installations can play a big role in reducing this financial drain. Considering that solar setup costs are gradually coming down, is this a good time to start investing in solar? To highlight true advantages of solar systems for academic institutions and corporates alike, we showcase here case studies of IIT Roorkee and Delhi International Airport.

Case 1: IIT Roorkee

The most important benefit of a

solar installation for institutions is that it reduces their grid energy consumption. IIT Roorkee is a good example, which has substantially cut its electricity expenses, LPG usage and carbon emission by investing in a solar setup.

The setup. IIT Roorkee used solar arrangements for two broad areas: departmental buildings, and hostels and residential areas. Solar PV systems across the roofs of its departmental buildings ensure minimum possible consumption of grid electricity. Hostels and residential areas use solar setups for cooking and water heating.

Major components of the complete arrangement are:

1. 27 units of a total 1.81MW polycrystalline solar photovoltaic system on departmental building roofs
2. A total of 435,000 litres per day (lpd) capacity of solar water heating (340,000lpd thermosiphon system and 90,000lpd flat-plate-bed system)
3. Solar steam cooking for nine hostel buildings using Scheffler concentrators
4. SCADA system for monitoring the total power generation and distribution

The results. By September 2016, the solar PV system generated net 5517.2MWp energy. This is enough to run the 27 departments of IIT-Roorkee covering a 33,278sq.m area, eliminating the need for grid electricity. Recent analytics indicate that

APPROXIMATE COSTS AND RETURNS OF IIT ROORKEE'S SOLAR ARRANGEMENT

Arrangements	Cost	Benefits
1.81MW solar PV system across 33,278sq.m area with 25-year maintenance contract	₹ 159 million	• Over 7000MWp electricity generation till July 2017, substantially decreasing utility electricity cost • 7208-tonne carbon dioxide reduction
440,000lpd solar water heating system with 20-year maintenance contract	₹ 119.3 million	2.6 million units electricity savings every year with a net 2132-tonne carbon footprint decrease
61 solar concentrators for cooking with 5-year maintenance	₹ 25.9 million	Energy generation replacing almost 5000 conventional 14.2kg LPG cylinders
MNRE subsidies for the setup	₹ 128.4 million for solar PV system ₹ 28.5 million for solar water heating ₹ 10.5 million for solar cooking setup	
Expected ROI	7-8 years	

Source: Prof. Arun Kumar, AHCC, IIT Roorkee